

Drawing Indicator Analysis

Distilled Knowledge Base for RAG

Extracted from 7 literature sources | Optimised for DIA RAG pipeline

PURPOSE: Contains only actionable drawing interpretation rules. All irrelevant content (neuroscience methodology, brain imaging research, art therapy process) has been removed. Use as the sole RAG knowledge source for DIA.

PART 1 - CORE INTERPRETATION PRINCIPLES

1.1 Non-Clinical Stance (Critical)

- Drawings must NEVER be used for clinical diagnosis. They are observational tools only.
- Use cautious phrasing: "may suggest" | "could reflect" | "possibly linked to" | "warrants attention" | "no strong concern evident"
- A single drawing is NEVER sufficient for conclusions - patterns across repeated drawings matter more.
- Never conflate artistic style choices with psychological indicators.
- Always consider developmental stage before flagging any indicator.
- If the child's verbal description contradicts a visual indicator, prioritise the child's words.

1.2 The Projection Principle

- Drawings are projections of the child's inner world. The drawing tells more about the drawer than the subject drawn.
- Drawings are direct communication from the unconscious - harder to camouflage than words.
- Approach drawings holistically first. The integrated whole is more than the sum of its parts.
- First impression matters: note overall emotional feeling before analysing individual elements.
- Even a tree drawing reflects the child's psychological state - all drawing content is expressive.

1.3 Repetition Rule

- Single symbols mean little. A theme appearing REPEATEDLY across drawings is significant.
- Analyse repetition of: colors, shapes, recurring details, themes, missing elements.
- Hasty conclusions from one drawing should always be avoided.

PART 2 - LINE PRESSURE AND QUALITY INDICATORS

2.1 Pen / Line Pressure

- Heavy pressure (High): Reflects muscle tension, high energy, emotional intensity. In boys especially common. Combined with other dark indicators may suggest anxiety or aggression.
- Light/Faint pressure (Low): Suggests low physical and mental energy, possible shame or depression-associated indicators.
- Variable pressure (Normal): Demonstrates flexibility and adaptability - a POSITIVE indicator.

2.2 Line Quality

- Excessive darkening of lines or figures is a concern signal when it appears repeatedly.
- Crossing out abnormally within a drawing warrants attention.
- Shapes that are consistently not closed may reflect incomplete emotional processing.

PART 3 - SHADING AND COLOR TONE INDICATORS

3.1 Shading - Body

- Whole body shaded heavily: Indicates generalised anxiety.
- Specific body part shaded: Anxiety associated with that body part or function.
- Heavy distorted shading (child distorts shape while shading): More common in anxious children who lack compatibility with their environment.

3.2 Overall Color Tone

- Dark, heavy colors overall: Linked to psychopathology indicators (Altschuler & Hattwick, 1947) - interpret cautiously, only when consistent.
- Varied, bright colors: Healthy children use wider color variety - POSITIVE indicator.
- Children do not choose colors randomly - color selection reveals mood and personality.

- Dark tone + corner placement + small figure = pattern worth noting as a combination.

IMPORTANT: Dark colors used *ONCE* are *NOT* a concern signal. Only flag when consistent across multiple drawings.

PART 4 - FIGURE SIZE AND PAGE USAGE

4.1 Figure Size

- Very large/exaggerated figures: Usually indicate aggression and overactivity. Children who do not adapt well tend to draw exaggerated figures.
- Very small/tiny figures: Suggests feelings of incompetence, shame, fear, or depressive indicators.
- Proportional size: Normal developmental indicator.

4.2 Page Usage and Placement

- Large page usage / big drawings: Reflects confidence, need for expressive space - can be POSITIVE.
- Tiny drawings confined to a corner: Concern signal when combined with dark tone and missing elements.
- Center placement: Developmental norm; balanced self-concept.
- Corner or side placement: Can reflect withdrawal, insecurity, or isolation indicators.
- COMBINATION SIGNAL: Corner placement + small figure + dark tone = strengthened concern signal.

PART 5 - HUMAN FIGURE BODY PART INDICATORS

5.1 Head and Face

- Exaggerated/very large head: Inflated ego indicator in self-drawings.
- Large eyes: Child feels watched or controlled by others.
- Small circles for eyes: Dependence and shallow emotion indicator.
- Absent/deleted eyes: Unwillingness to engage with others - significant concern signal.
- Very large mouth with large teeth: Aggression, verbal aggression, desire to dominate.
- Enlarged/emphasised nostrils: Aggression indicator.
- Absent facial features entirely (no face at all): Significant concern signal - refusal to express emotion or engage.

5.2 Neck

- Too long neck: Difficulties achieving desires and satisfying needs.
- Absent neck: Emotional disconnection between thought (head) and action (body).

5.3 Hands - Key Indicator

- Outstretched hands: Desire to connect, willingness to help and interact - POSITIVE indicator.
- Small hands: Feelings of insecurity and helplessness.
- Missing hands: One of the most significant concern signals. Seen in anxious, helpless, or abused children.

Missing hands is among the most literature-supported concern indicators. Always flag when absent from human figures.

5.4 Arms and Legs

- Missing arms: Passive withdrawal from engagement.
- Missing legs: Instability, difficulty standing ground.
- Missing body parts generally: Anxiety about, or denial of, those body functions.

PART 6 - SOCIAL AND RELATIONAL INDICATORS

6.1 Figures and Relationships

- Child describing happy family but drawing only one tiny figure: Disconnection between verbal and visual expression.
- Absence of figures entirely in a child who typically draws people: Social withdrawal indicator.
- Consistent refusal to draw people at all: Can reflect trauma.

6.2 Figure Distance and Self Positioning

- Figures placed far apart: Emotional distance in relationships; possible isolation or loneliness.
- Figures placed close together: Emotional closeness; sense of security - POSITIVE indicator.
- Self drawn with others: Sense of belonging - POSITIVE.

- Self drawn separately or in corner, away from group: Possible isolation or rejection indicator.
- Self drawn significantly smaller than other figures: Feelings of inferiority or submission.

PART 7 - CONSOLIDATED CONCERN SIGNALS AND POSITIVE INDICATORS

7.1 Signals That WARRANT ATTENTION (from literature)

These patterns, especially when recurring, warrant gentle non-diagnostic attention.

- Recurring overall feeling of unease from the drawing.
- Certain body parts (especially hands/face) always missing.
- Eyes that are squinting, hollow, or absent.
- Abnormal or excessive crossing out within the drawing.
- Excessive darkening throughout the drawing.
- Drawings consistently tiny or confined to a small corner.
- Child consistently refuses to draw or describe drawings.
- Same drawing theme repeated within a short period.
- Figures are entirely empty (no features).
- Shapes consistently unclosed.
- Self figure separated from all others, especially with corner placement.

7.2 Signals That Are NOT Automatically Concerning

- Violent scenes: NOT automatically worrying unless repeated obsessively.
- Dark colors used once: NOT a concern signal without repetition.
- Unrealistic or abstract drawing style: NOT a deficit.
- Missing body parts in children under 7: Often developmentally NORMAL (tadpole drawings).
- Large page usage: Often a POSITIVE indicator of confidence.

7.3 POSITIVE Emotional Indicators - Must Be Reported

DIA must identify and report positive indicators, not only concerns. These are literature-supported positive signals.

- Bright, varied color palette: Positive emotional expression and engagement.
- Large figures drawn with confidence: Healthy self-concept and self-assertion.
- Figures smiling or with positive facial expressions: Happiness indicator.
- Figures holding hands or positioned closely: Positive attachment and relational indicators.
- Sun, flowers, houses in drawing: Security, warmth, and stability symbols. House especially significant in ages 6-7.
- Outstretched arms on figures: Desire to connect - POSITIVE.
- Varied colorful scenes with multiple figures: Healthy, engaged emotional expression.
- Large page usage with confident composition: Positive expressive confidence.
- Variable line pressure: Flexibility and adaptability.
- Close figure placement: Emotional warmth and connection.

PART 8 - COLOR PSYCHOLOGY

8.1 Color Principles

- Children do not choose colors randomly - selection reveals mood and personality.
- Overall color tone (dark, balanced, light) is a key first-pass indicator of emotional atmosphere.
- Heavy use of black/dark throughout: Possible anxiety or depression indicators - only when consistent.
- Heavy use of red: Can indicate aggression or high arousal, but also love and excitement - context matters.
- Varied, bright palette: Positive engagement - healthy children use more color variety.
- Monochromatic palette (all one color): May reflect emotional constriction or fixation if consistent.

PART 9 - INTEGRATION AND INTERPRETATION RULES

9.1 Interpretation Priority Order

- 1. Child's own verbal/written description overrides visual indicators if they contradict.

- 2. Repeated patterns across multiple drawings override single-instance observations.
- 3. Combinations of multiple indicators carry more weight than isolated single indicators.
- 4. Developmental stage must be considered before any interpretation.
- 5. Cultural context must be considered - Western realism norms do not apply universally.

9.2 When NOT to Flag Concern

- If only ONE indicator is present in isolation.
- If the child's verbal description strongly contradicts the visual indicator.
- If the indicator can be explained by developmental stage.
- If it is a one-time occurrence not supported by repetition.
- If no concern: explicitly state "no strong psychological concern is evident from the observable indicators."

9.3 When to Note Concern

- Two or more concern indicators co-occurring.
- The child's verbal description aligns with the concerning visual indicators.
- The pattern is inconsistent with the child's developmental stage.
- Specific body parts (especially hands/face) consistently missing.
- Extreme isolation of self-figure combined with dark tone and small figure size.

PART 10 - DEVELOPMENTAL AND CULTURAL CONTEXT

10.1 Age-Stage Awareness

- Ages 4-7: Tadpole figures, missing body parts, simplified forms are developmentally NORMAL. Do not over-pathologise.
- Ages 8-12: Children strive for realism. Rigidity and stereotyping are NORMAL at this stage.
- Always reference developmental stage before flagging indicators.

10.2 Cultural Context

- Cultural symbolism affects how children draw (Alland, 1983). Same mental representation may be expressed differently across cultures.
- Do NOT impose Western cultural norms of realistic drawing as the standard.

APPENDIX - HYPERPARAMETER RECOMMENDATIONS FOR RAG PIPELINE

Based on distillation of 7 sources into this single clean knowledge file, the following parameters are recommended for the DIA RAG pipeline.

Recommended Settings

top_k = 6 (was 4)

Allows coverage of: visual indicators, integration rules, core principles, positive indicators, social indicators, and contextual rules. Current value of 4 is too restrictive for balanced two-way interpretation.

MAX_CONTEXT_CHARS = 5500 (was 3500)

At top_k=6 and 5500 chars, each chunk gets ~917 chars - sufficient for meaningful content per chunk.

chunk_size = 750 (was 900)

Shorter focused chunks produce better semantic retrieval matches with this clean, high-signal source document.

overlap = 120 (was 150)

Slightly reduced - less noise with cleaner source material.

REPLACE all 7 original PDFs with this single distilled document. This removes approximately 60% noise content while retaining 100% of actionable DIA interpretation rules. The retrieval query remains unchanged - no code changes needed beyond swapping the data folder.